FPGA Lab 0

Cameron Thomas

## Csynth latency

```


================================================================
== Vitis HLS Report for 'top_kernel'
================================================================
* Date:           Sun Jan 18 13:13:25 2026

* Version:        2025.1.1 (Build 6214317 on Sep 11 2025)
* Project:        project_1
* Solution:       hls (Vivado IP Flow Target)
* Product family: zynquplus
* Target device:  xczu3eg-sbva484-1-e


================================================================
== Performance Estimates
================================================================
+ Timing:
    * Summary:
    +--------+----------+----------+------------+
    |  Clock |  Target  | Estimated| Uncertainty|
    +--------+----------+----------+------------+
    |ap_clk  |  10.00 ns|  7.300 ns|     2.70 ns|
    +--------+----------+----------+------------+

+ Latency:
    * Summary:
    +---------+---------+----------+----------+-----+-----+-------------------------------------------------+
    |  Latency (cycles) |  Latency (absolute) |  Interval |                     Pipeline                    |
    |   min   |   max   |    min   |    max   | min | max |                       Type                      |
    +---------+---------+----------+----------+-----+-----+-------------------------------------------------+
    |      117|      117|  1.170 us|  1.170 us|  100|  100|  loop auto-rewind stp (delay=0 clock cycles(s))|
    +---------+---------+----------+----------+-----+-----+-------------------------------------------------+

```

## Cosim latency and clock period

```
Report time        : Sun Jan 18 13:14:17 EST 2026.
Solution           : hls.
Simulation tool    : xsim.

+----------+----------+-----------------------------------------------+-----------------------------------------------+----------------------+
|          |          |             Latency(Clock Cycles)             |             Interval(Clock Cycles)            | Total Execution Time |
+   RTL    +  Status  +-----------------------------------------------+-----------------------------------------------+    (Clock Cycles)    +
|          |          |      min      |      avg      |      max      |      min      |      avg      |      max      |                      |
+----------+----------+-----------------------------------------------+-----------------------------------------------+----------------------+
|      VHDL|        NA|             NA|             NA|             NA|             NA|             NA|             NA|                    NA|
|   Verilog|      Pass|            140|            140|            140|             NA|             NA|             NA|                   140|
+----------+----------+-----------------------------------------------+-----------------------------------------------+----------------------+

```

Post implementation resource utilization

```


Implementation tool: Xilinx Vivado v.2025.1.1
Project:             project_1
Solution:            hls
Device target:       xczu3eg-sbva484-1-e
Report date:         Sun Jan 18 13:22:49 EST 2026

#=== Post-Implementation Resource usage ===
SLICE:            0
LUT:           1906
FF:            3019
DSP:              3
BRAM:             2
URAM:             0
LATCH:            0
SRL:            216
CLB:            472

#=== Final timing ===
CP required:                     10.000
CP achieved post-synthesis:      2.779
CP achieved post-implementation: 2.845
Timing met

```